

Design of a Reconfigurable Approximate 8-Bit Multiplier for Energy-Aware Applications

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Abstract—This paper presents a reconfigurable 8-bit approximate multiplier architecture that enables dynamic accuracy–efficiency trade-offs through a mode-select control. The proposed design integrates lightweight approximation techniques, including selective partial-product reduction and simplified carry paths, to achieve reduced computational energy and power dissipation in low-accuracy mode while preserving high-fidelity multiplication in accurate mode. By allowing real-time switching between operating modes, the multiplier adapts to the error-tolerance characteristics of different workloads such as image processing, signal analytics, and edge-AI inference.

Index Terms—Approximate computing, approximate multiplier, reconfigurable multiplication, mode-select architecture, error-tolerant multiplier.

I. INTRODUCTION

The growing demand for high-performance, energy-efficient digital systems has intensified research into arithmetic circuit optimisation. Domains such as image processing, machine learning, signal processing, and Internet-of-Things (IoT) devices often tolerate small computational inaccuracies in exchange for significant reductions in power consumption, latency, and hardware area. Conventional 8-bit array and tree multipliers incur substantial switching activity due to full partial-product generation and long carry-propagation paths. Approximate computing provides a means to relax an exact configurable and precision-tunable arithmetic unit by introducing controlled inaccuracies, thereby reducing logic complexity and dynamic power in error-tolerant applications.

However, most approximate multiplier architectures employ static approximations, preventing them from adapting to varying precision requirements within multi-stage signal-processing pipelines. To address this limitation, this work introduces a reconfigurable 8-bit approximate multiplier featuring a mode-select signal that dynamically configures partial-product reduction and carry-propagation logic. The accurate mode preserves exact arithmetic, while the approximate mode activates simplified computation paths that reduce hardware transitions and critical-path delay.

This tunable architecture enables application-dependent quality scaling, making it suitable for edge-AI workloads, lightweight Digital Signal Processing applications, and battery-constrained embedded systems requiring flexible accuracy–efficiency trade-offs.

II. RELATED WORKS

Approximate arithmetic has been widely explored as a means of achieving higher performance and better energy efficiency in error-resilient applications, including image processing, neural inference, and low-power signal processing. Previous approximate multipliers can be broadly divided into three categories: truncation-based, compressor-based, and adder-based designs. Truncation-oriented approaches discard part of the partial products or low-significance columns to reduce logic depth and switching activities. In compressor-based designs, modifications to 3:2 or 4:2 compressor cells inside the partial-product reduction trees form an approximate arithmetic block with a shortened critical path. These approximations are bound to multi-stage compression; most circuits also have limited flexibility after synthesis. Adder-based approximate multipliers focus on simplifying the accumulation stage by replacing accurate adders with error-resilient or carry-truncation units. Representative examples of Lower-Part OR Adder (LOA), Error-Tolerant Adders (ETA), AMA, and related derivatives have reported significant delay and power gains at the expense of degraded numerical accuracy.

Reconfigurable and hybrid approximate multipliers have been proposed as ways to support adjustable precision, but many strategies like these typically involve significant structural modifications to the datapath, and complicate timing closure and verification. Fine-grained adaptability has also been explored more recently through mode-selective approximate adders; most of those architectures have nonetheless continued to use traditional compressor-tree-based reduction networks.

The two main differences between the proposed design and the previous work are as follows: (a) the complete partial-product matrix is kept intact, same as the Wallace tree structure. Thus, all approximations take place in a Multi-Mode Adder(MMA) block. (b) The proposed MMA has mode-dependent operand usage; that is, it processes only three partial-product rows in the full-accuracy mode, whereas it uses all four of them in the other approximate modes. Therefore, it facilitates runtime control of latency and accuracy without modification to the partial product generation stage. This adder-based and uniform architecture provides more complexity and higher regularity, with a more predictable error behaviour across modes, compared to existing methodologies for approximate multipliers.

III. PROPOSED DESIGN

A. Partial product generation

The proposed multiplier uses a classic 8×8 partial-product generation structure: all 64 products $P_{ij} = A_i B_j$ are generated using simple AND gates. These partial products are organized into a weighted matrix based on their bit positions to ensure full compatibility with exact operation when the accurate mode is enabled. No approximation is introduced during this step to keep the datapath structurally identical in both modes to verify the process and ensure correctness during the accurate configuration.

B. Accumulation stage-I

Partial-product matrix accumulation follows a multistage compression approach, followed by a final summation. All columns with multiple partial products are reduced using exact or approximate 3:2 or 4:2 compressors, depending on the mode of operation. During the accurate mode, all compressors propagate carries normally, acting like a standard array multiplier. In the approximate mode, the compressors within the low-significance region suppress or truncate low-significance input signals to reduce the logic depth and switching activity.

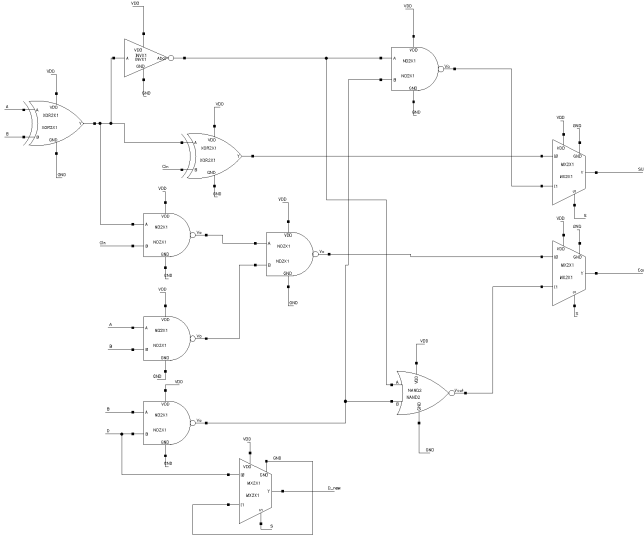


Fig. 1. MultiMode Adder circuit

Finally, the two intermediate sum rows obtained are fed to the proposed MMA for final addition. The MMA maintains full carry propagation during accurate mode, while it uses simplified logic in the lower bits during the approximate mode to reduce the computational complexity. The structured accumulation naturally localizes approximation only to the carry-sensitive portions of the datapath; hence, bounded and predictable error characteristics are maintained while full accuracy is preserved whenever required.

During every accumulation step, four rows of PP are fed as inputs to the MMA, and it operates according to the

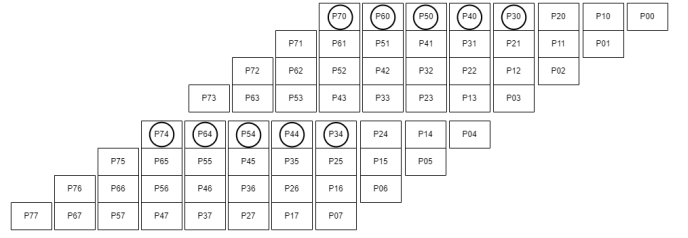


Fig. 2. Accumulation stage-I of the proposed multiplier [4]

accuracy mode chosen. In the accurate mode, MMA performs full multi-operand addition using only the three most significant bits out of four PP inputs, whereas the fourth PP row is cleanly bypassed and passed onto the subsequent accumulation stage. This guarantees that the accumulated sum exhibits behaviour of exact arithmetic with full carry propagation, ensuring correctness similar to any traditional multirow reduction. During approximate modes, all four PP inputs are processed simultaneously by the MMA, allowing higher parallelism at different stages. The MMA suppresses or truncates the carry propagation in the lower-significance region and minimizes the critical path and switching activity, producing the combined output within the same stage instead of forwarding an additional PP row. This mode-dependent treatment of PP inputs at run-time allows the architecture to dynamically increase the accumulation throughput by inducing controlled, localized inaccuracy in a tunable speed–accuracy trade-off. [4] From figure 2, the bigger rectangle with four sub-blocks represents the Multi-mode adder and the sub-block marked with a circle is the least significant PP. Similarly, rectangles with three and two sub-blocks are traditional FAs and HAs, respectively.

C. Accumulation Stage - II

The outputs of the Accumulation Stage-I are fed to Stage-II [5] with another row of multiple MMAs. It further enhances the energy efficiency at the cost of accuracy. In a similar way to Stage-I, SUM outputs of Stage-I and COUT of Stage-I are fed as the four inputs of the Stage-II MMAs. Similarly, SUM, remaining single PP and COUT are fed to Stage-II FAs and HAs. The outputs are forwarded to Stage-III.

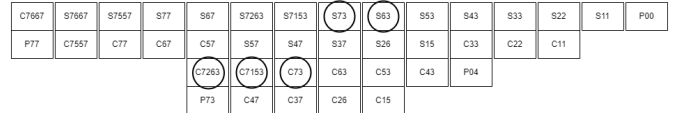


Fig. 3. Accumulation stage-II of the proposed multiplier

D. Approximate Multiplication (Stage-III)

This stage computes the approximate result of the multiplication. The outputs of Stage-II are fed to this stage and are processed serially to get the approximate result. For this, a series of FAs and HAs are used.

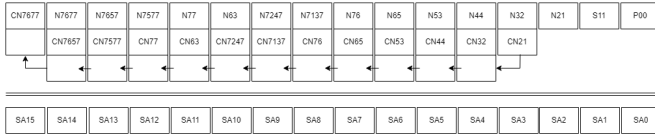


Fig. 4. Stage-III of the approximate proposed multiplier

In the figure 4, the arrows represent the carry propagating serially from one adder to the next. The output thus obtained is the final approximate multiplication result ($SA_{15} - SA_0$)

E. Error Correction Stage

When the multiplier is operating in exact mode, all the less significant bypassed bits by the MMA are collected and the error bit sequence is formed by addition.

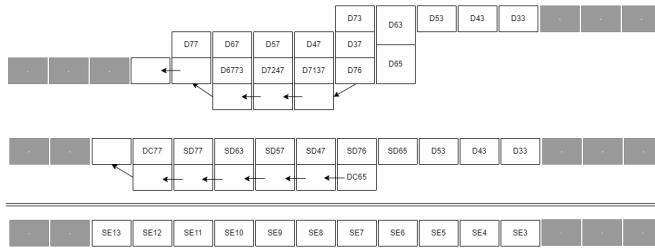


Fig. 5. Error Sequence generation blocks

In the figure 5, the first block shows the bypassed bits generated in Accumulation Stage-I and II. The outputs are fed in the next block and finally the error sequence ($SE_{13} - SE_3$) are generated.

F. Exact Multiplication (final stage)

When the MODE is set to exact mode ($5'b00000$), the error bits generated else the error sequence will be set to 0. Thus, the error sequence, when added to the approximate result with FAs and HAs, will yield the exact multiplication result ($S_{15} - S_0$).

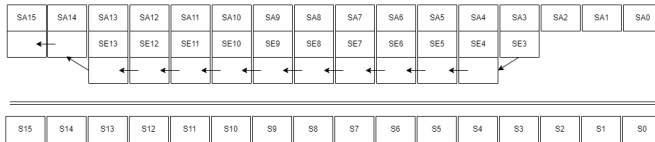


Fig. 6. Exact Multiplication calculation

G. Parallel-In-Parallel-Out Stage (PIPO)

Different output bits will reach at different times due to different path delays. Therefore, to ensure that all the bits reach the output at same instant of time, sixteen flip-flops in PIPO are intentionally placed at the output. Subsequently, care should be taken that the flip-flop clock cycle is more than the maximum path data path delay of the entire multiplier.

$$t_{clk} > t_{path,max} \quad (1)$$

MODE	Input
MODE 0	00000
MODE 1	00001
MODE 2	00011
MODE 3	00111
MODE 4	01111
MODE 5	11111

Trade-offs of this proposed design

The proposed design prioritises energy efficiency at the cost of certain trade-offs:

- The area of the design is not optimized and is much more than the traditional array architecture.
- Extra sub-blocks are added for configurable accuracy, which increases the complexity of the design.
- Although the goal is to reduce the delay, the gain is not as significant as expected.

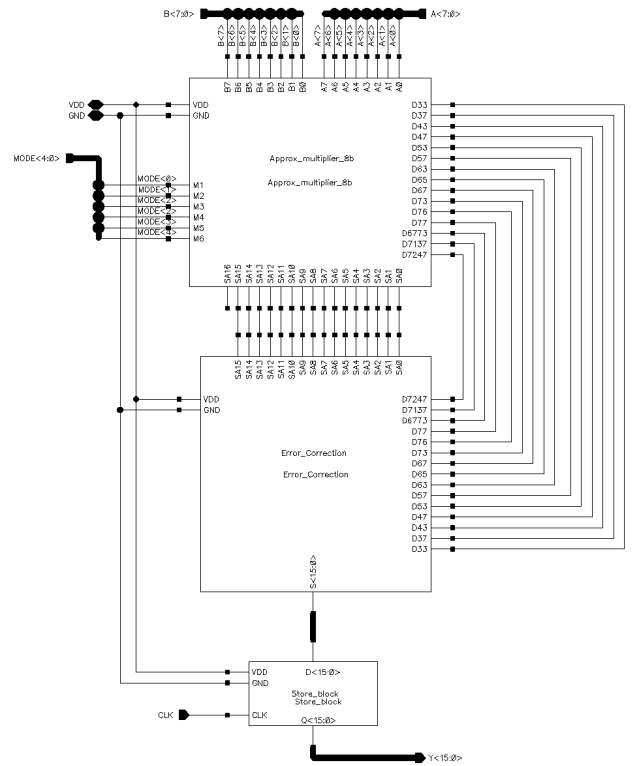


Fig. 7. Proposed Architecture of 8 X 8 Configurable Accuracy Approximate Multiplier

IV. RESULTS AND DISCUSSIONS

To evaluate the performance of the configurable accuracy multiplier, 55 sets of inputs were taken, and the results were evaluated across the six modes of accuracy and analyzed. For critical path delay calculation, we have to follow the path from A_7 to S_{15} .

A. Accuracy Analysis

[4] To analyse the accuracy of the multiplier, various error metrics were evaluated. Assuming that the approximate value of an output is ϕ_i , the exact corresponding one is ω_i , and the absolute value of the difference between them is the error distance (ED). At present, the evaluation metrics that are extensively applied include the mean error distance (MED), the normalized mean error distance (NMED), the mean relative error distance (MRED), the mean squared error (MSE), the number of effective bits (NoEB), and the worst-case error (WCE). These metrics are calculated as follows: [1] [6] (Common error metrics)

$$ED_i = |\phi_i - \omega_i| \quad (2)$$

$$MED = \frac{\sum_{i=1}^{2n} ED_i}{2^{2n}} \quad (3)$$

$$NMED = \frac{MED}{(2^n - 1)^2} \quad (4)$$

$$RED_i = \frac{ED_i}{\omega_i} \quad (\forall \omega_i \neq 0) \quad (5)$$

$$MRED = \frac{\sum_{i=1}^{2n} MRED_i}{2^{2n}} \quad (6)$$

$$MSE = \frac{\sum_{i=1}^{2n} ED_i^2}{2^{2n}} \quad (7)$$

$$NoEB = 2n - \log_2(1 + \sqrt{MSE}) \quad (8)$$

$$WCE = MAX(ED_i) \quad (9)$$

(Error influencing factors)

$$ER = \frac{count(ED_i \neq 0)}{2^{2n}} \quad (10)$$

$$TED = \sum_{t=1}^{ER \cdot 2^{2n}} ED_t \quad (11)$$

By these equations, the error metrics were calculated

TABLE I
ERROR METRICS [2] [3]

Metrics	M0	M1	M2	M3	M4	M5
MED	0	13.236	74.909	596.727	1.554K	2.988K
NMED	0	0.204m	1.152m	9.176m	23.91m	45.96m
MRED	0	0.00327	0.0125	0.0752	0.147	0.233
MSE	0	403.78	10.066K	0.633M	3.702M	17M
NoEB	16	11.601	9.337	6.362	5.089	3.990
WCE	0	48	240	2288	4848	10992
ER%	0	58.18	80	90.90	96.36	96.36
TED	0	728	4120	32820	85517	164376

From figure 8, the percentage error is increasing with each mode as more significant bits are approximated [6]. For

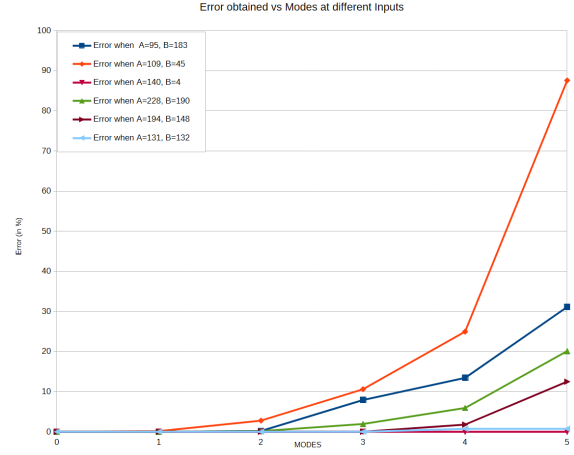


Fig. 8. Error obtained variations at different modes for some different combinations of inputs

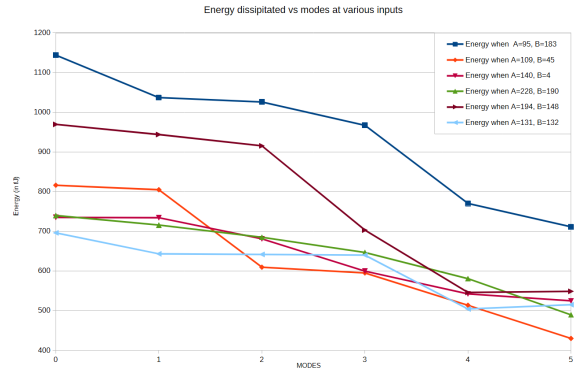


Fig. 9. Energy dissipation at different modes for some different combinations of inputs

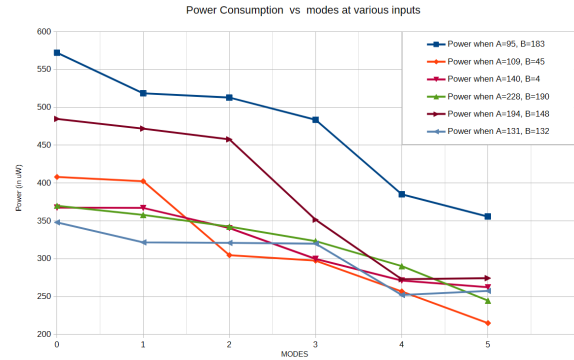


Fig. 10. Power consumption at different modes for some different combinations of inputs [3]

various inputs, energy efficiency is observed in figures 9 and 10. Delay is not observed to drop significantly in the various modes in figure 11. Various FoMs are plotted (Figures 12-15) where the reduced accuracy is seen to bring about gains in Energy and Power dissipation.

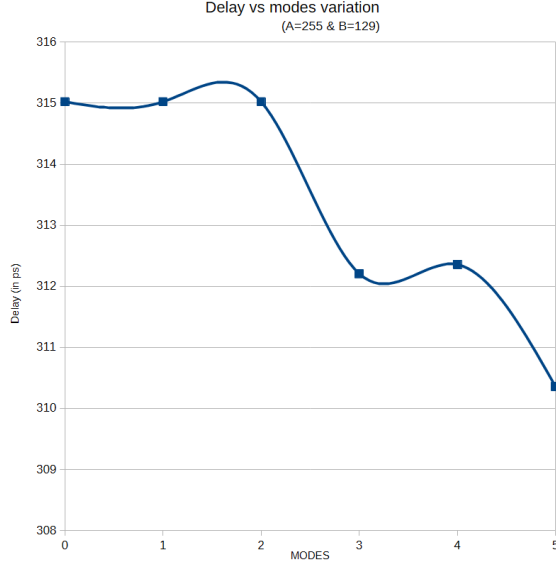


Fig. 11. Delay at different modes for some different combinations of inputs

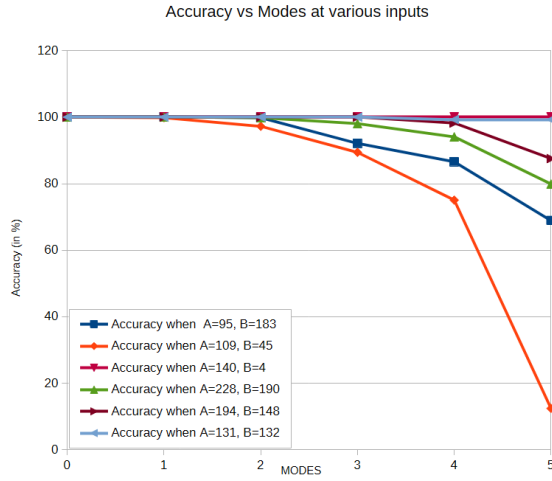


Fig. 12. Accuracy variation at different modes

V. CONCLUSION

In this work, we introduced an 8-bit approximate multiplier architecture that leverages a Multimode Adder (MMA) to provide tunable accuracy–performance trade-offs without altering the partial product generation process. Traditional approximate multipliers often integrate approximation at several points in the datapath—such as partial product perforation, aggressive compression-tree modification, or speculative carry manipulation—resulting in complex control logic and irregular timing behavior. In contrast, the proposed design intentionally preserves a conventional partial product matrix and shifts all configurability to the accumulation stage. This approach simplifies hardware design while still enabling significant flexibility in operation.

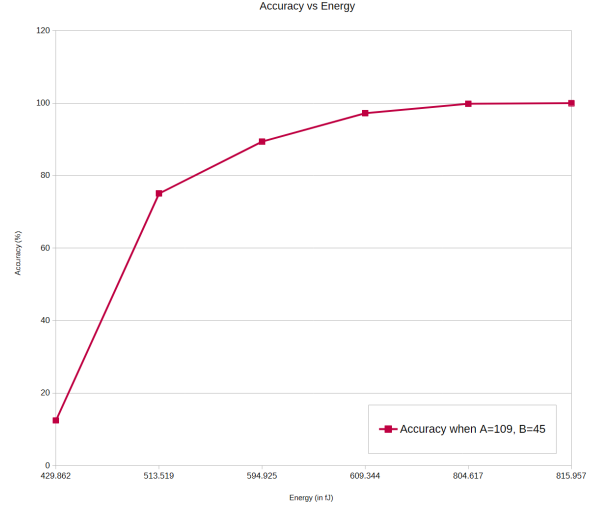


Fig. 13. Accuracy variation w.r.t energy dissipation

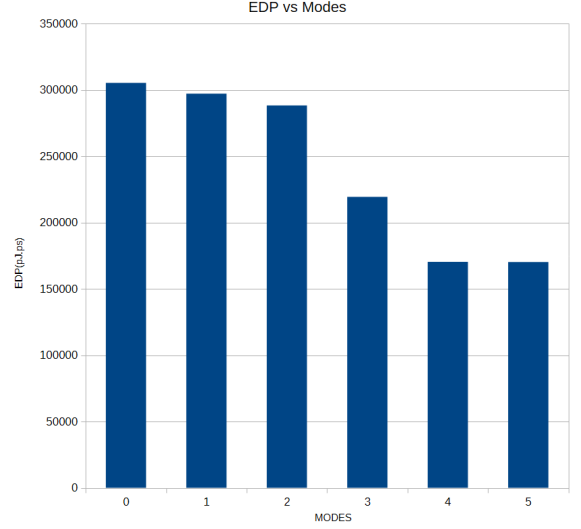


Fig. 14. EDP variation at different modes [4]

The MMA serves as the core of the architecture, enabling the multiplier to seamlessly switch between accurate and approximate modes. In accurate operation, the MMA accumulates three partial products while bypassing the fourth for subsequent processing, thereby preserving full-precision behavior. In approximate modes, all four partial products are combined directly within the MMA, allowing the design to relax carry propagation paths and shorten the critical path. This selective use of partial products within a unified adder structure not only reduces computation time but also decreases overall resource utilization, making the architecture particularly suitable for lightweight, low-power systems.

By centralizing approximation within a single stage, the proposed method offers several benefits. First, the design maintains predictable timing characteristics, as the partial product

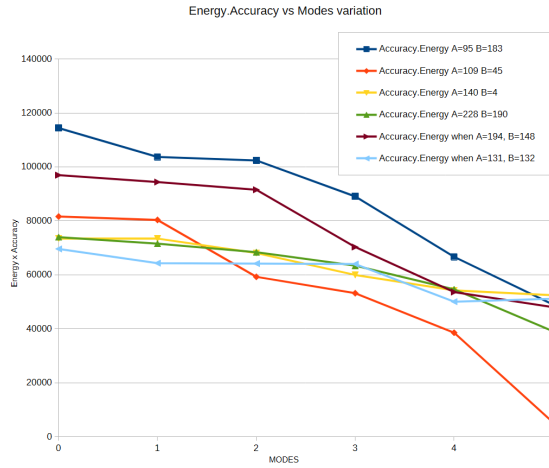


Fig. 15. Energy x Accuracy variation at different modes for different inputs

network remains unmodified across all modes. Second, the reconfigurable MMA introduces minimal control and wiring overhead compared to multi-stage approximation pipelines. Third, the architecture can be easily scaled or integrated into larger arithmetic units, signal-processing datapaths, or machine-learning accelerators where dynamic precision scaling is essential.

Overall, the presented architecture demonstrates that substantial efficiency improvements can be achieved through localized approximation rather than global datapath restructuring. The proposed multimode accumulation strategy provides a clean balance between accuracy, delay, and hardware cost while remaining suitable for both general-purpose and application-specific designs. Future extensions may include exploring adaptive mode selection driven by workload characteristics, expanding the MMA concept to wider multipliers or fused multiply-accumulate units, and incorporating approximate adder families with error-tunable carry chains to further broaden the design space of configurable arithmetic architectures.

During the project, several key observations emerged. A primary takeaway was that it is unnecessary to approximate every component of the circuit; selecting a single stage for reconfigurability can effectively balance hardware efficiency with meaningful performance-accuracy trade-offs. The work also deepened understanding of partial product structures and their influence on timing characteristics, error behavior, and the formation of the critical path.

The development of the MMA required designing hardware blocks capable of supporting multiple operational modes without introducing excessive overhead or compromising correctness. Overall, the project strengthened knowledge of approximate arithmetic techniques, modular hardware design principles, and the practical design compromises necessary for creating efficient, scalable multiplier architectures.

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